

Lecture 3:

Natural Deduction

Proofs in Propositional Logic

- Truth tables quickly become unusable: A proposition with n atoms (variables) has 2^n possible assignments to those atoms.
- Instead we want to prove things directly.
- Here we'll use a system called *natural deduction*.

Introduction and Elimination Rules

- Every connective will have ways to *introduce* them and ways to *eliminate* them.
- Elimination rules are generally used to break down the premises, while introduction rules get us towards our goal.
- For instance to prove $Q \wedge R \rightarrow P \vee Q \vee X$ we would presumably use an elimination rule to extract Q from $Q \wedge R$ and two introduction rules to derive $P \vee Q \vee X$, plus a rule to introduce our arrow.

AND Rules

$$\frac{P \quad Q}{P \wedge Q} \wedge\text{-I}$$

$$\frac{P \wedge Q}{P} \wedge\text{-E-L}$$

$$\frac{P \wedge Q}{Q} \wedge\text{-E-R}$$

OR Rules

$$\frac{P}{P \vee Q} \text{ V-I-L}$$

$$\frac{Q}{P \vee Q} \text{ V-I-R}$$

$$\frac{P \vee Q \quad P \rightarrow R \quad Q \rightarrow R}{R} \text{ V-E}$$

IF-THEN Rules

$$\frac{\begin{array}{c} \overline{P}^A \\ \hline \vdots \\ \hline Q \end{array}}{P \rightarrow Q} \rightarrow\text{-I}$$

$$\frac{P \quad P \rightarrow Q}{Q} \rightarrow\text{-E}$$

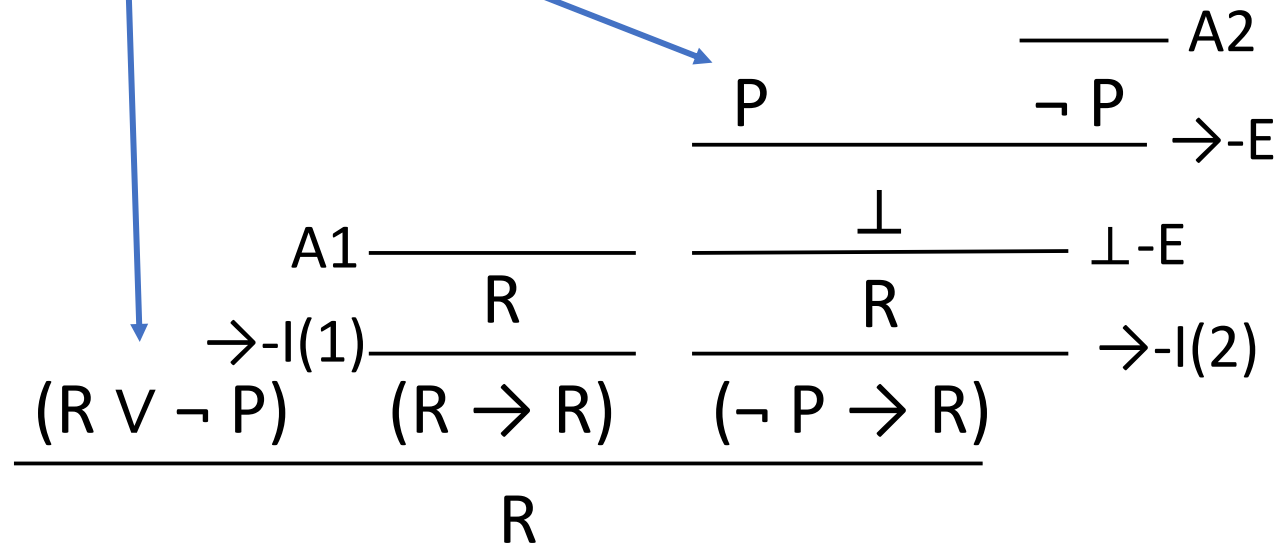
NOT

$\neg P := P \rightarrow \perp$

$$\frac{\perp}{P} \quad \neg\text{-E}$$

Using False

- Claim: If P and $(R \vee \neg P)$ then R



Other formalisms

$$P, P \rightarrow Q \vdash Q$$
$$\frac{\begin{array}{l} P \\ P \rightarrow Q \end{array}}{\therefore Q}$$

1. P	A
2. $P \rightarrow Q$	A
3. Q	MPP (1,2)